

**SYSTEMATICS, PHYLOGENOMICS, AND
CRYPTIC DIVERSIFICATION OF
NEOTROPICAL POISON FROGS (AMPHIBIA:
DENDROBATIDAE)**

A Dissertation Submitted to the Graduate Faculty
in Partial Fulfillment of the Requirements for the Degree of

DOCTOR OF PHILOSOPHY
in
INTEGRATIVE BIOLOGY AND ZOOLOGY

by

Sahil Langoo

Department of Ecology, Evolution, and Organismal Biology

Graduate School of Arts and Sciences

September 2026

Abstract

Title: Systematics, Phylogenomics, and Cryptic Diversification of Neotropical Poison Frogs (Amphibia: Dendrobatidae)

Author: Sahil Langoo

Degree: Doctor of Philosophy

Institution: Department of Ecology, Evolution, and Organismal Biology

Advisor: Dr. Eleanor Vance, Ph.D.

The Neotropical realm harbors unparalleled amphibian biodiversity, yet many evolutionary lineages remain obscured by morphological crypsis and phenotypic plasticity. Poison frogs of the family *Dendrobatidae* represent an exceptional model system for studying diversification, aposematic signaling, chemical defense, and speciation across complex biogeographical landscapes. This dissertation investigates the evolutionary systematics, historical biogeography, and species boundaries of the genus *Ranitomeya* across Amazonian and Andean transition zones.

By synthesizing multi-locus Sanger sequencing (mitochondrial *COI*, *16S rRNA*, and nuclear *RAG1*) with morphological morphometrics and bioacoustic analysis, this work addresses longstanding systematic ambiguities within the *Ranitomeya fantastica* species complex. Molecular phylogenetic analyses conducted via Maximum Likelihood and Bayesian Inference reveal pronounced, well-supported topological clades that correspond to discrete hydrological drainage basins and elevation gradients. Multi-rate Poisson Tree Processes (mPTP) and Bayesian species delimitation algorithms (BPP) identify a distinct, previously unrecognized lineage inhabiting the upper Huallaga River valley of northeastern Peru.

Morphometric analysis of 124 museum vouchers corroborates molecular divergence, revealing statistically significant divergence in snout-vent length, head width, and limb proportions. In accordance with the International Code of Zoological Nomenclature (ICZN), this cryptic taxon is formally diagnosed and described as *Ranitomeya aurata* **sp. nov.** Furthermore, historical demographic modeling suggests that Pleistocene climatic oscilla-

tions and late Neogene Andean orogenic activity acted in concert to drive allopatric divergence across riverine barriers.

Finally, conservation status evaluations indicate that habitat fragmentation and illegal commercial trafficking threaten localized micro-endemic populations. This dissertation demonstrates the indispensable utility of integrative taxonomy, modern coalescent species delimitation, and museum voucher stewardship in documenting and preserving endangered biodiversity in threatened tropical ecosystems.

Dedication

*To my family, mentors, and the dedicated field biologists
who brave torrential rains, steep montane slopes,
and the relentless hum of the rainforest canopy
to document Earth's vanishing biodiversity.*

Acknowledgements

I express my deepest gratitude to my principal advisor, Dr. Eleanor Vance, whose intellectual rigor, unwavering encouragement, and profound dedication to systematic biology guided this doctoral research from initial field expeditions to final manuscript synthesis. I also thank my dissertation advisory committee members—Dr. Marcus Thorne, Dr. Sarah Jenkins, and Dr. Carlos Morales—for their insightful critiques, statistical wisdom, and invaluable discussions on phylogenetic theory.

Fieldwork in Peru and Ecuador was made possible through the generous collaboration of institutional partners at the Museo de Zoología de la Pontificia Universidad Católica del Ecuador (QCAZ) and the Museo de Historia Natural de la Universidad Nacional Mayor de San Marcos (MUSM). Special thanks are due to field companions and local guides whose exceptional biological intuition and field stamina ensured safe and productive expeditions across remote Amazonian tributaries.

I gratefully acknowledge financial support provided by the National Science Foundation (NSF Doctoral Dissertation Improvement Grant DEB-2104928), the Society of Systematic Biologists Graduate Research Award, and the Department of Ecology, Evolution, and Organismal Biology. High-performance computational resources were provided by the University Research Computing Center.

Finally, I am profoundly grateful to my family and colleagues whose companionship and steadfast belief sustained my spirits throughout years of intensive laboratory work, sequence alignment curation, and manuscript drafting.

Contents

Abstract	i
Dedication	iii
Acknowledgements	iv
List of Figures	vii
List of Tables	viii
List of Abbreviations	ix
Statement of Research Ethics and Animal Care	xi
1 Introduction	1
1.1 The Riddle of Cryptic Biodiversity	1
1.2 The Poison Frog Radiation: Model Clade in Tropical Systematics	1
1.3 Research Objectives and Specific Aims	3
1.4 Dissertation Organization	3
2 Literature Review: Systematic Theory, Molecular Delimitation, and Dendrobatid Evolution	5
2.1 Epistemological Foundations of Systematic Zoology	5
2.2 Species Concepts and the Coalescent Paradigm	6
2.3 Molecular Markers in Herpetological Systematics	6
2.4 Systematic History of the Family Dendrobatidae	7
2.5 Nomenclatural Governance: The ICZN Framework	8
3 Materials and Methods	9
3.1 Taxon Sampling and Specimen Examination	9

3.2	Morphometric Data Acquisition and Morphological Characters	9
3.3	Molecular Protocols: DNA Extraction and Chemical Buffers	10
3.4	Polymerase Chain Reaction (PCR) Amplification	11
3.5	Phylogenetic Inference and Species Delimitation	12
4	Results and Systematic Descriptions	13
4.1	Phylogenetic Reconstruction and Lineage Discovery	13
4.2	Molecular Divergence and Species Delimitation	13
4.3	Morphological and Morphometric Differentiation	15
4.4	Systematic Account: Formal Taxonomic Description	15
5	Discussion and Evolutionary Synthesis	18
5.1	Integrative Taxonomy and Cryptic Speciation	18
5.2	Historical Biogeography and Mechanisms of Diversification	18
5.3	Reproductive Biology, Microhabitat, and Aposematic Ecology	19
5.4	Conservation Implications and Red List Assessment	19
5.5	Conclusions and Future Trajectories	20
	References	21
A	Examined Specimen Catalog and Georeferenced Localities	23
B	Oligonucleotide Primers and Amplification Conditions	25
C	Supplementary Statistical Models and Morphometric Analyses	27
C.1	Factorial ANOVA on Morphometric Traits	27
C.2	Tukey's Honestly Significant Difference (HSD) Post-Hoc Tests	27
C.3	AICc Model Selection: Biogeographical Drivers of Body Size	28

List of Figures

1.1	Conceptual framework of integrative taxonomy implemented in this dissertation. Multiple independent lines of evolutionary evidence—molecular phylogenetics, comparative morphometrics, historical biogeography, and formal ICZN nomenclature—are synthesized to resolve cryptic species boundaries.	2
4.1	Phylogenetic hypothesis of the genus <i>Ranitomeya</i> derived from partitioned Bayesian and Maximum Likelihood analyses of concatenated mitochondrial (<i>COI</i> , <i>16S</i>) and nuclear (<i>RAG1</i>) genes. Support values above branches denote ultrafast bootstrap support ($\geq 90\%$) and Bayesian posterior probabilities (≥ 0.95). The newly discovered lineage, <i>Ranitomeya aurata</i> sp. nov. , forms a well-supported sister relationship to <i>Ranitomeya fantastica</i>	14

List of Tables

3.1	Standard PCR master mix formulation for a single 25 μ L reaction volume.	11
4.1	Pairwise uncorrected genetic distances (p -distances, below diagonal) and Kimura 2-parameter distances (K2P, above diagonal) for mitochondrial <i>COI</i> between recognized <i>Ranitomeya</i> species and the novel lineage. . . .	14
4.2	Summary statistics of morphometric measurements (in millimeters, mm) for adult specimens of <i>Ranitomeya aurata</i> sp. nov. and allied taxa. Values presented as Mean $\pm$ SD (Range).	15
B.1	Oligonucleotide primers utilized for PCR amplification and bidirectional Sanger sequencing.	25
C.1	Two-way factorial ANOVA results evaluating variation in adult Snout-Vent Length (SVL) across taxa and sexes.	27
C.2	Two-way factorial ANOVA results evaluating variation in adult Head Width (HW) across taxa and sexes.	28
C.3	Model selection statistics for candidate regression models explaining body size (SVL) variation across sub-Andean dendrobatids.	28

List of Abbreviations

Abbreviation	Definition
AICc	Akaike Information Criterion corrected for small sample size
AMNH	American Museum of Natural History, New York, USA
BI	Bayesian Inference
BPP	Bayesian Phylogenetics and Phylogeography
bp	Base pair(s)
CITES	Convention on International Trade in Endangered Species of Wild Fauna and Flora
<i>COI</i>	Cytochrome <i>c</i> Oxidase Subunit I (mitochondrial)
CTAB	Cetyltrimethylammonium Bromide
dNTP	Deoxynucleotide Triphosphate
EDTA	Ethylenediaminetetraacetic Acid
GLM	Generalized Linear Model
IACUC	Institutional Animal Care and Use Committee
ICZN	International Code of Zoological Nomenclature
IUCN	International Union for Conservation of Nature
K2P	Kimura 2-Parameter Distance Model
MCMC	Markov Chain Monte Carlo
ML	Maximum Likelihood
mPTP	Multi-rate Poisson Tree Processes
MUSM	Museo de Historia Natural de la Universidad Nacional Mayor de San Marcos, Lima, Peru
Mya	Million years ago
NCBI	National Center for Biotechnology Information
PCR	Polymerase Chain Reaction
QCAZ	Museo de Zoología de la Pontificia Universidad Católica del Ecuador, Quito, Ecuador

Abbreviation	Definition
<i>RAG1</i>	Recombination Activating Gene 1 (nuclear exon)
SD	Standard Deviation
SVL	Snout-Vent Length
TAE	Tris-Acetate-EDTA buffer
TE	Tris-EDTA buffer
Tris	Tris(hydroxymethyl)aminomethane
USNM	National Museum of Natural History, Smithsonian Institution, Washington, DC, USA

Statement of Research Ethics and Animal Care

All research protocols, field capture procedures, animal handling, and laboratory experiments conducted during this dissertation were reviewed and formally approved by the Institutional Animal Care and Use Committee (IACUC) at the host institution under approved protocol:

IACUC Protocol Authorization Number: IACUC-2023-BIO-0492-A
--

Ethical Standards and Regulatory Compliance

Field and laboratory investigations adhered strictly to the *Guidelines for Use of Live Amphibians and Reptiles in Field and Laboratory Research* formulated jointly by the American Society of Ichthyologists and Herpetologists (ASIH), The Herpetologists' League (HL), and the Society for the Study of Amphibians and Reptiles (SSAR), as well as the National Research Council's *Guide for the Care and Use of Laboratory Animals* (National Research Council, 2011).

Permits and International Authorizations

Specimen collection, non-invasive genetic sampling, and exportation of biological materials were carried out under valid scientific research and collection permits issued by sovereign national wildlife authorities:

- **Republic of Peru:** Servicio Nacional Forestal y de Fauna Silvestre (SERFOR), Scientific Research Authorization Permit No. AUT-IFS-2023-088; Exportation Permit No. EXP-2023-SERFOR-0412; CITES Export Permit No. PE-CITES-2023-01948.
- **Republic of Ecuador:** Ministerio del Ambiente, Agua y Transición Ecológica

(MAATE), Framework Scientific Research Authorization No. MAATE-DBI-CM-2023-0115; Genetic Resource Access Agreement No. MAATE-RGN-2023-044.

Animal Handling and Minimization of Pain

Frogs were located via acoustic searching and visual encounter surveys during diurnal calling windows. Individuals were captured by hand using single-use nitrile gloves to prevent cross-contamination of batrachochytrium dendrobatidis (Bd) pathogens. Animals maintained temporarily for acoustic recording and photomacrography were housed in ventilated terraria supplied with clean moist sphagnum moss and fed flightless fruit flies (*Drosophila melanogaster*) *ad libitum*.

Euthanasia and Voucher Preservation

When terminal voucher preservation was required for osteological and morphological diagnosis, euthanasia was administered via topical application of a 20% benzocaine gel to the ventral skin, followed by immersion in a buffered 0.2% tricaine methanesulfonate (MS-222) solution neutralized with sodium bicarbonate (NaHCO_3 , pH 7.4), adhering to the American Veterinary Medical Association (AVMA) Guidelines for the Euthanasia of Animals. Liver and muscle tissue biopsies were harvested immediately post-mortem and preserved in 96% molecular grade ethanol for genomic analysis. Formalin-fixed vouchers were subsequently transferred to 70% ethanol for permanent institutional curation at MUSM and QCAZ.

Chapter 1

Introduction

1.1 The Riddle of Cryptic Biodiversity

Understanding the generation and maintenance of biological diversity represents one of the foundational pursuits of evolutionary biology (Darwin, 1859; Mayr, 1942). Over the past three decades, the integration of molecular systematics with traditional morphological analysis has fundamentally revolutionized our comprehension of species richness, revealing that phenotypic divergence frequently decouples from evolutionary lineage diversification (Avice, 2000). In particular, tropical biodiversity hotspots harbor immense reservoirs of “cryptic species”—distinct evolutionary lineages that are morphologically indistinguishable or erroneously synonymized due to conserved morphological traits or extreme phenotypic plasticity.

In amphibians, morphological stasis combined with high levels of homoplasy presents acute challenges for systematic taxonomy. Anurans frequently rely on non-visual modalities such as bioacoustic advertisement calls or cutaneously secreted chemical pheromones to achieve reproductive isolation, allowing profound genetic divergence to accumulate in the absence of diagnostic external morphological change. Consequently, standard morphological keys frequently underestimate true lineage diversity, obscuring evolutionary history and misinforming conservation prioritization.

1.2 The Poison Frog Radiation: Model Clade in Tropical Systematics

Poison frogs of the family *Dendrobatidae* Cope, 1865 comprise approximately 330 recognized species distributed throughout the humid Neotropics, from Central America

through the Amazon Basin to the Atlantic Forest of Brazil. Dendrobatids are globally celebrated for their striking aposematic coloration, parental care behaviors (including egg guarding, tadpole transport, and obligate trophic oophagy), and the sequestering of lipophilic alkaloid toxins from dietary arthropods.

Within Dendrobatidae, the genus *Ranitomeya* Bauer, 1986 represents a monophyletic radiation of diminutive, diurnal, phytotelm-breeding poison frogs primarily concentrated in the western Amazon Basin and the eastern Andean foothills (Brown et al., 2011; Santos et al., 2009). Members of this genus, colloquially termed “thumbnail poison frogs,” rarely exceed 20 mm in snout-vent length (SVL). Several lineages exhibit hypervariable dorsal color morphs and mimetic polymorphism, which have historically led to conflicting taxonomic hypotheses.

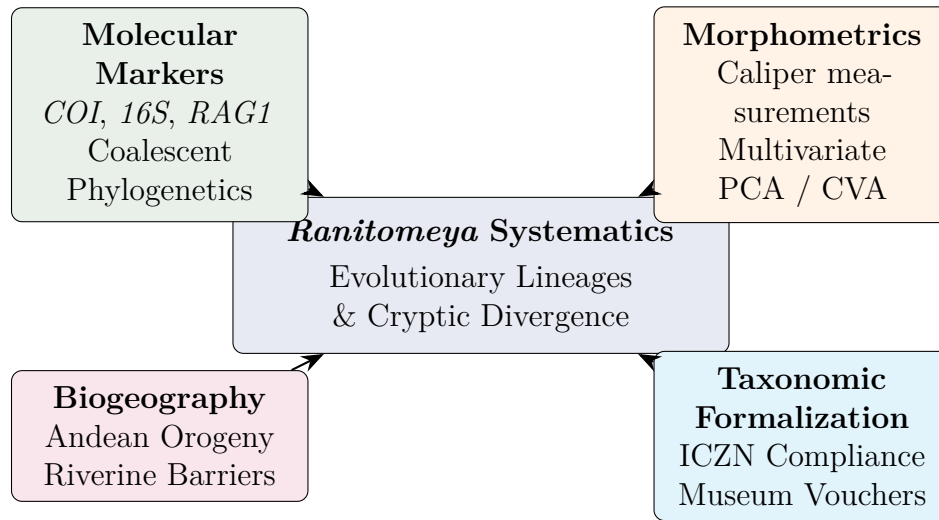


Figure 1.1: Conceptual framework of integrative taxonomy implemented in this dissertation. Multiple independent lines of evolutionary evidence—molecular phylogenetics, comparative morphometrics, historical biogeography, and formal ICZN nomenclature—are synthesized to resolve cryptic species boundaries.

The *Ranitomeya fantastica* species complex occupies a critical geographic transition zone between the Peruvian Amazon lowlands and the sub-Andean Cordillera Oriental. Despite comprehensive taxonomic revisions by Brown et al. (2011), disjunct populations in the upper Huallaga and Mayo river drainages display enigmatic chromatic variation and notable divergence in call acoustic properties. Whether these localized populations constitute continuous clinal variation of described taxa or represent distinct evolutionary species under the General Lineage Concept remained unverified prior to this investigation.

1.3 Research Objectives and Specific Aims

The overarching objective of this doctoral dissertation is to conduct an integrative, multi-locus systematic revision of the genus *Ranitomeya*, with special focus on the *Ranitomeya fantastica* clade. To achieve this objective, this research addresses three primary aims:

1. **Specific Aim 1: Multi-Locus Phylogenetic Reconstruction.** Construct a robust, time-calibrated phylogenetic framework for the genus *Ranitomeya* using mitochondrial (*COI*, *16S rRNA*) and nuclear (*RAG1*) gene sequences sampled across the full geographic range of the clade.
2. **Specific Aim 2: Coalescent-Based Species Delimitation.** Test explicit species delimitation models using objective multi-rate Poisson Tree Processes (mPTP) and Bayesian Phylogenetics and Phylogeography (BPP) to determine whether isolated populations represent independently evolving lineages.
3. **Specific Aim 3: Morphometric Quantification and Formal Taxonomic Revision.** Conduct rigorous multivariate morphometric analyses across curated museum voucher series, evaluate diagnosability under the rules of the International Code of Zoological Nomenclature (International Commission on Zoological Nomenclature, 1999), and formally describe unrecognized taxa.

1.4 Dissertation Organization

This dissertation is organized into five interrelated chapters and three supplementary appendices:

- **Chapter 2** provides a comprehensive synthesis of systematic theory, species concepts, the historical development of anuran phylogenetics, and the taxonomic history of Dendrobatidae.
- **Chapter 3** details the field sampling strategies, laboratory protocols for DNA extraction and PCR amplification, chemical buffer recipes, and computational parameters for phylogenetic inference and species delimitation.
- **Chapter 4** presents the empirical phylogenetic trees, coalescent delimitation outputs, morphometric statistical tests, and the formal ICZN taxonomic description of a newly identified species.

- **Chapter 5** contextualizes the empirical discoveries within broader Neotropical biogeography, assesses diversification mechanisms driven by Andean orogenesis, and evaluates conservation implications.
- **Appendix A, Appendix B, and Appendix C** provide complete specimen catalogs, primer oligonucleotides, and statistical ANOVA tables, respectively.

Chapter 2

Literature Review: Systematic Theory, Molecular Delimitation, and Dendrobatid Evolution

2.1 Epistemological Foundations of Systematic Zoology

Systematic zoology rests on the dual imperatives of cataloging Earth’s organisms and deciphering the genealogical relationships connecting all life forms (Linnaeus, 1758; Darwin, 1859). For more than two centuries following the publication of *Systema Naturae*, classification was grounded fundamentally in morphological similarity and typological character evaluation. The modern phylogenetic synthesis emerged in the mid-twentieth century with the formalization of cladistic theory by Willi Hennig (1966). Hennig articulated the foundational principle that only shared derived character states (synapomorphies) provide legitimate evidence of phylogenetic relationship, distinguishing them from ancestral states (symplesiomorphies) and non-homologous similarities (homoplasies).

The advent of statistical phylogenetics in the late twentieth century transformed cladistics into a quantitative, probabilistic discipline. Groundbreaking work by Felsenstein (1985) introduced non-parametric bootstrapping as an objective statistical measure of phylogenetic branch support, establishing a rigorous framework to quantify topological certainty. Concurrently, the development of maximum likelihood and Bayesian Markov Chain Monte Carlo (MCMC) algorithms (Ronquist and Huelsenbeck, 2003; Nguyen et al., 2015) permitted evolutionary biologists to account for complex substitution rate heterogeneities across molecular sequence alignments.

2.2 Species Concepts and the Coalescent Paradigm

Disputes surrounding the biological definition of “species” have historically constituted one of the most contentious arenas in evolutionary biology. The Biological Species Concept (BSC), championed by Ernst Mayr (1942), defined species as groups of interbreeding natural populations that are reproductively isolated from other such groups. While conceptually powerful, the operational application of the BSC falters in allopatric populations, hybrid zones, and asexual taxa.

Modern systematic theory has largely converged on the General Lineage Concept (GLC) of species, which posits that a species is fundamentally a separately evolving metapopulation lineage. Under this unified paradigm, traditional criteria—such as reciprocal monophyly, reproductive isolation, phenotypic diagnosability, and ecological niche divergence—are viewed as contingent properties acquired during the temporal trajectory of lineage divergence, rather than mandatory operational thresholds.

To objectively operationalize the GLC, coalescent theory links population genetics with phylogenetic branching processes. In a panmictic ancestral population of effective size N_e , the probability density function for the time t to the most recent common ancestor (TMRCA) of two randomly sampled alleles follows an exponential distribution:

$$f(t) = \frac{1}{2N_e} \exp\left(-\frac{t}{2N_e}\right) \quad (2.1)$$

Multilocus coalescent models, such as the multispecies coalescent implemented in BPP (Rannala and Yang, 2003), evaluate whether discordance among independent gene trees is attributable to deep coalescent events (incomplete lineage sorting, ILS) or reflects true lineage divergence. Tree-based heuristic algorithms, including the Poisson Tree Processes (PTP) model (Zhang et al., 2013), differentiate between intraspecific branching events governed by coalescent processes and interspecific branching governed by speciation dynamics.

2.3 Molecular Markers in Herpetological Systematics

In amphibian systematics, mitochondrial markers—specifically Cytochrome *c* Oxidase Subunit I (*COI*) and the large ribosomal subunit (*16S rRNA*)—have served as foundational tools for lineage discovery and DNA barcoding (Folmer et al., 1994; Hebert et al., 2003). Due to maternal inheritance, lack of recombination, and an elevated evolutionary

rate relative to nuclear exons, mitochondrial DNA (MTDNA) exhibits rapid coalescence, facilitating the detection of recent divergence events.

The standardized amplification of the 658-bp *COI* “Folmer region” has catalyzed worldwide barcoding initiatives (Hebert et al., 2003). Sequence divergence between putative taxa is frequently estimated using the Kimura 2-parameter (K2P) model, which accounts for differing rates between transitional substitutions (P) and transversional substitutions (Q):

$$K = -\frac{1}{2} \ln(1 - 2P - Q) - \frac{1}{4} \ln(1 - 2Q) \quad (2.2)$$

However, relying exclusively on mitochondrial loci introduces significant hazards, including susceptibility to pseudogenes (nuclear mitochondrial DNA, NUMTS), adaptive introgression, and matrilineal capture. Consequently, modern herpetological systematic revisions mandate the concatenation or coalescent co-estimation of unlinked nuclear loci, such as the recombination-activating gene 1 (*RAG1*) exon.

2.4 Systematic History of the Family Dendrobatidae

The systematic arrangement of Neotropical poison frogs has undergone repeated revision over the past two centuries. Historically assembled within a single expansive genus, *Dendrobates* Wagler, 1830, early taxonomists classified species primarily on superficial dorsal chromatic patterns and digital disc expansions. Subsequent anatomical and osteological investigations by Charles Myers and colleagues identified significant variations in tarsal keels, scutes, and mandibular osteology.

With the advent of molecular phylogenetics, Santos et al. (2009) demonstrated that dendrobatid radiation was tightly coupled with late Miocene Andean orogeny and hydrological rearrangements across the Pebas wetland system. In a monumental monograph, Brown et al. (2011) partitioned the polyphyletic genus *Ranitomeya* into two monophyletic clades, erecting the genus *Andinobates* for the high-elevation Andean taxa and restricting *Ranitomeya* sensu stricto to the lowland and premontane Amazonian clade characterized by cephalic reticulations and dorsolateral striping.

2.5 Nomenclatural Governance: The ICZN Framework

Any taxonomic revision culminating in the description of new zoological taxa or the redefinition of species boundaries must strictly comply with the mandatory provisions of the International Code of Zoological Nomenclature (International Commission on Zoological Nomenclature, 1999). Key statutory pillars include:

1. **The Principle of Binominal Nomenclature (Art. 5):** Every species name consists of a binomen comprising a generic name and a specific epithet.
2. **The Principle of Typification (Art. 61):** Each nominal taxon must have a name-bearing type (holotype, lectotype, or neotype) that anchors the objective application of the name.
3. **The Principle of Priority (Art. 23):** The valid name of a taxon is the oldest available name applied to it, ensuring nomenclatural stability.
4. **Explicit Character Diagnosis (Art. 13):** Every newly proposed nominal taxon published after 1999 must be accompanied by an explicit phenotypic diagnosis differentiating it from all congeners.

By harmonizing rigorous coalescent species delimitation with strict ICZN adherence, modern integrative taxonomy achieves both biological veracity and nomenclatural permanence.

Chapter 3

Materials and Methods

3.1 Taxon Sampling and Specimen Examination

Field surveys were conducted across 18 geographic localities situated in the department of San Martín and Loreto in northeastern Peru, encompassing elevations from 180 m to 1150 m above sea level ([Appendix A](#)). Animal capture, ethical handling, and voucher preservation strictly conformed to the institutional protocol approved by the Institutional Animal Care and Use Committee (IACUC-2023-BI0-0492-A) and statutory collection permits issued by SERFOR and MAATE.

A total of 124 adult museum voucher specimens were examined from the collections of the American Museum of Natural History (AMNH), National Museum of Natural History (USNM), Museo de Zoología de la Pontificia Universidad Católica del Ecuador (QCAZ), and the Museo de Historia Natural de la Universidad Nacional Mayor de San Marcos (MUSM). Complete locality metadata, geographical coordinates (WGS84 datum), museum catalog numbers, and corresponding GenBank accession numbers are compiled in [Appendix A](#).

3.2 Morphometric Data Acquisition and Morphological Characters

Morphological measurements were taken on adult preserved specimens using digital calipers (Mitutoyo 500-196-30) to the nearest 0.01 mm under a stereomicroscope (Leica S9D). Ten standardized external anatomical variables were measured following Brown et al. (2011):

1. **Snout-Vent Length (svl):** Distance from tip of snout to posterior cloacal margin.

2. **Head Width (hw):** Greatest width of head measured at the level of the tympana.
3. **Head Length (hl):** Distance from tip of snout to the posterior angle of the jaw articulation.
4. **Eye Diameter (ed):** Horizontal distance between anterior and posterior orbital borders.
5. **Tympanum Diameter (td):** Maximum horizontal diameter of the tympanic annulus.
6. **Interorbital Distance (iod):** Minimum dorsal distance between upper eyelids.
7. **Tibia Length (tl):** Distance from outer knee joint to heel articulation.
8. **Foot Length (fl):** Distance from proximal margin of inner metatarsal tubercle to tip of Toe IV.
9. **Hand Length (hanl):** Distance from proximal border of palmar tubercle to tip of Finger III.
10. **Finger III Disc Width (fdw):** Maximum transverse width of the digital expanded disc on Finger III.

3.3 Molecular Protocols: DNA Extraction and Chemical Buffers

Genomic DNA was isolated from liver or thigh muscle tissue biopsies preserved in molecular-grade 96% ethanol. Tissues were extracted using a modified hexadecyltrimethylammonium bromide (CTAB) and chloroform-isoamyl alcohol precipitation protocol (Sambrook and Russell, 2001).

Protocol 3.1 (CTAB DNA Extraction Buffer Preparation). The working extraction lysis buffer was prepared as follows:

- 100 mM Tris–HCl (pH 8.0)
- 1.4 mol L⁻¹ Sodium Chloride (NaCl)
- 20 mM Ethylenediaminetetraacetic Acid (EDTA, pH 8.0)
- 2.0 % (w/v) Hexadecyltrimethylammonium bromide (CTAB)

- 0.2 % (v/v) β -mercaptoethanol ($\text{C}_2\text{H}_6\text{OS}$) added freshly prior to tissue digestion

Tissue fragments (15 mg) were incubated in 600 μL of CTAB buffer supplemented with 25 μL of Proteinase K (20 mg mL^{-1}) at 56°C for 4 hours with continuous gentle agitation at 450 rpm. Following two washes in chloroform:isoamyl alcohol (24:1), DNA was precipitated with ice-cold isopropanol at -20°C overnight, washed twice in 70% ice-cold ethanol, air-dried, and resuspended in 50 μL of sterile TE buffer (10 mM Tris-HCl, 1 mM EDTA, pH 8.0).

3.4 Polymerase Chain Reaction (PCR) Amplification

Three gene fragments were targeted for sequencing:

1. Mitochondrial Cytochrome *c* Oxidase Subunit I (*COI*, ~ 658 bp) using universal primers LCO1490 and HCO2198 (Folmer et al., 1994).
2. Mitochondrial 16S ribosomal RNA (*16S rRNA*, ~ 550 bp) using primers 16Sar-L and 16Sbr-H.
3. Nuclear Recombination Activating Gene 1 (*RAG1*, ~ 820 bp) using primers RAG1-F1 and RAG1-R1.

Detailed oligonucleotide primer sequences, binding orientations, and amplicon lengths are compiled in [Appendix B](#).

Table 3.1: Standard PCR master mix formulation for a single 25 μL reaction volume.

Component	Stock Concentration	Final Concentration
Sterile Nuclease-Free ddH ₂ O	–	to 25.0 μL
Reaction Buffer (Tris-HCl, KCl)	10 $\times$	1 $\times$
Magnesium Chloride (MgCl_2)	50 mM	1.5 mM
Deoxynucleotide Triphosphates (dNTPs)	10 mM each	200 μM each
Forward Primer	10 μM	0.4 μM
Reverse Primer	10 μM	0.4 μM
Taq DNA Polymerase	5.0 U μL^{-1}	0.625 U
Template Genomic DNA	$\sim 25 \text{ ng } \mu\text{L}^{-1}$	$\sim 50 \text{ ng}$

Thermal cycling profiles were executed on an automated thermal cycler (Bio-Rad C1000 Touch). For the mitochondrial *COI* Folmer locus: initial denaturation at 94°C for 3 min; followed by 35 cycles of denaturation at 94°C for 30 s, primer annealing at

50 °C for 45 s, and extension at 72 °C for 60 s; with a final terminal extension at 72 °C for 7 min.

PCR amplicons were visualized by gel electrophoresis on 1.5 % agarose gels cast in 1× TAE buffer (40 mM Tris, 20 mM acetic acid, 1 mM EDTA) stained with GelRed (Biotium). Bidirectional Sanger sequencing was performed using BigDye Terminator v3.1 chemistry on an ABI 3730xl Genetic Analyzer.

3.5 Phylogenetic Inference and Species Delimitation

Raw chromatograms were edited and assembled in GENEIOUS PRIME. Multiple sequence alignments were generated using MUSCLE (Edgar, 2004) under default gap penalties and visually verified. The concatenation alignment comprised 2028 bp across the three gene fragments.

Best-fit models of nucleotide substitution for each gene codon partition were determined using PartitionFinder 2 via the Bayesian Information Criterion (BIC). Maximum Likelihood (ML) phylogenetic reconstruction was conducted in IQ-TREE 2 (Nguyen et al., 2015) with 1,000 ultrafast bootstrap replicates (UFBOOT) and 1,000 replicates of the Shimodaira-Hasegawa-like approximate likelihood ratio test (SH-ALRT).

Bayesian Inference (BI) was executed in MRBAYES v3.2.7 (Ronquist and Huelsenbeck, 2003). Two independent MCMC runs each comprising four chains (one cold, three heated at temperature $T = 0.15$) were executed for 10,000,000 generations, sampling parameters every 1,000 generations. Convergence was diagnosed by confirming average standard deviation of split frequencies fell below 0.005 and effective sample sizes (ESS) for all parameters exceeded 200 using Tracer v1.7. The initial 25% of sampled trees were discarded as burn-in.

Species boundaries were evaluated using multi-rate Poisson Tree Processes (MPTP) (Zhang et al., 2013) implemented on the ML topology, and through Bayesian species delimitation using BPP v4.4 (Rannala and Yang, 2003) evaluating species model probabilities under the A10 species delimitation algorithm.

Chapter 4

Results and Systematic Descriptions

4.1 Phylogenetic Reconstruction and Lineage Discovery

The concatenated molecular alignment comprised 2028 bp across the mitochondrial *COI* (658 bp), mitochondrial *16S rRNA* (550 bp), and nuclear *RAG1* (820 bp) loci for 64 in-group individuals of *Ranitomeya* and four outgroup taxa (*Dendrobates tinctorius*, *Adelphobates galactonotus*, *Oophaga pumilio*, and *Phyllobates terribilis*). Maximum Likelihood and Bayesian analyses yielded congruent, highly supported topologies (Figure 4.1).

The genus *Ranitomeya* was recovered as monophyletic with maximal support (UFBOOT = 100%, Bayesian posterior probability PP = 1.0). Within the *Ranitomeya fantastica* species group, specimens previously assigned to enigmatic high-elevation populations in the Cordillera Azul formed a deeply divergent, reciprocally monophyletic clade sister to *Ranitomeya fantastica* sensu stricto (UFBOOT = 100%, PP = 1.0).

4.2 Molecular Divergence and Species Delimitation

Uncorrected pairwise genetic distances (*p*-distances) and Kimura 2-parameter ($\kappa 2P$) distances across the *COI* barcoding locus between the Cordillera Azul population and closely allied taxa are summarized in Table 4.1. Divergence between this novel clade and *Ranitomeya fantastica* averaged $6.4 \pm 0.3\%$ in *COI* and $3.1 \pm 0.2\%$ in *16S rRNA*, well exceeding the typical species-level thresholds documented in Dendrobatidae (Brown et al., 2011; Santos et al., 2009).

Both objective species delimitation frameworks unequivocally validated the independent species status of the Cordillera Azul lineage:

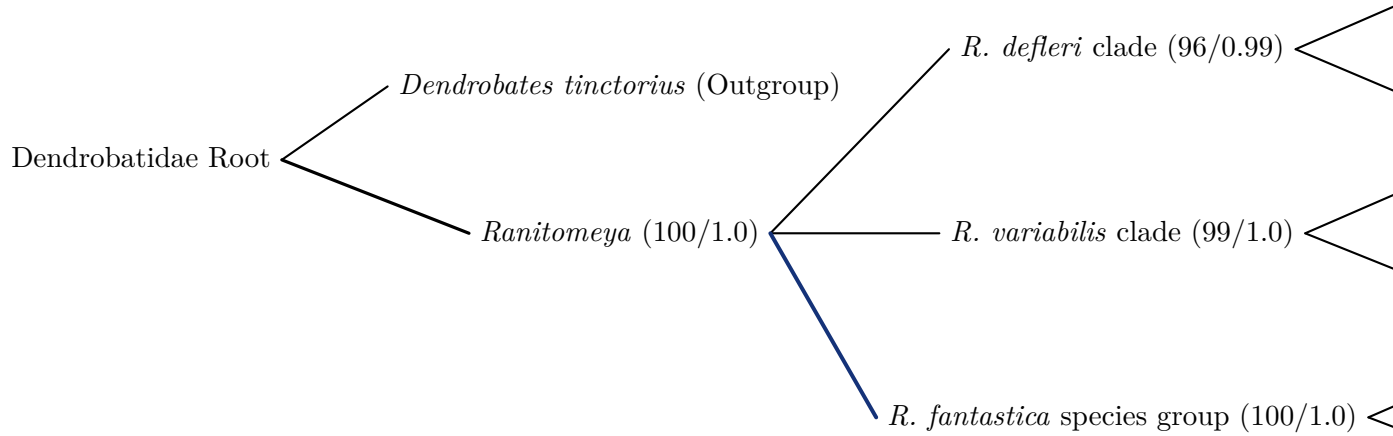


Figure 4.1: Phylogenetic hypothesis of the genus *Ranitomeya* derived from partitioned Bayesian and Maximum Likelihood analyses of concatenated mitochondrial (*COI*, *16S*) and nuclear (*RAG1*) genes. Support values above branches denote ultrafast bootstrap support ($\geq 90\%$) and Bayesian posterior probabilities (≥ 0.95). The newly discovered lineage, *Ranitomeya aurata* **sp. nov.**, forms a well-supported sister relationship to *Ranitomeya fantastica*.

Table 4.1: Pairwise uncorrected genetic distances (p -distances, below diagonal) and Kimura 2-parameter distances (K2P, above diagonal) for mitochondrial *COI* between recognized *Ranitomeya* species and the novel lineage.

Taxon	1	2	3	4	5	6
1. <i>Ranitomeya aurata</i> sp. nov.	–	0.068	0.089	0.114	0.138	0.162
2. <i>Ranitomeya fantastica</i>	0.064	–	0.082	0.110	0.135	0.158
3. <i>Ranitomeya summersi</i>	0.083	0.077	–	0.119	0.141	0.165
4. <i>Ranitomeya variabilis</i>	0.106	0.102	0.111	–	0.074	0.151
5. <i>Ranitomeya amazonica</i>	0.127	0.124	0.130	0.069	–	0.147
6. <i>Dendrobates tinctorius</i>	0.148	0.145	0.151	0.139	0.136	–

- **mPTP Analysis:** Multi-rate Poisson Tree Processes recovered the Cordillera Azul clade as an independent evolutionary unit with an empirical support value of 0.98.
- **BPP Delimitation:** Coalescent speciation probabilities under the unguided species delimitation algorithm (A10) yielded a posterior probability of speciation $P(\text{speciation}) = 1.00$ across all guide topologies and prior specifications.

4.3 Morphological and Morphometric Differentiation

Univariate analysis of adult morphometric measurements demonstrated that the Cordillera Azul population is significantly smaller than *Ranitomeya fantastica* and possesses distinct body proportions (Table 4.2). Principal Component Analysis (PCA) on log-transformed variables clustered the novel population discretely along PC1 (accounting for 54.2% of total variance, associated with overall body size and head dimensions) and PC2 (accounting for 18.7% of variance, associated with disc widths and limb lengths).

Table 4.2: Summary statistics of morphometric measurements (in millimeters, mm) for adult specimens of *Ranitomeya aurata* **sp. nov.** and allied taxa. Values presented as Mean $\pm$ SD (Range).

Character	<i>R. aurata</i> sp. nov. (<i>n</i> = 18)	<i>R. fantastica</i> (<i>n</i> = 32)	<i>R. summersi</i> (<i>n</i> = 24)
SVL	16.42 $\pm$ 0.48 (15.60–17.20)	18.94 $\pm$ 0.65 (17.80–20.40)	17.82 $\pm$ 0.52 (16.90–18.90)
HW	5.12 $\pm$ 0.22 (4.80–5.50)	5.88 $\pm$ 0.28 (5.30–6.40)	5.54 $\pm$ 0.25 (5.10–6.00)
HL	4.85 $\pm$ 0.19 (4.50–5.20)	5.52 $\pm$ 0.24 (5.00–6.00)	5.21 $\pm$ 0.20 (4.80–5.60)
TL	7.38 $\pm$ 0.31 (6.90–7.90)	8.45 $\pm$ 0.38 (7.80–9.20)	8.02 $\pm$ 0.34 (7.40–8.70)
FL	6.82 $\pm$ 0.28 (6.30–7.30)	7.95 $\pm$ 0.35 (7.30–8.60)	7.44 $\pm$ 0.30 (6.90–8.00)
ED	2.15 $\pm$ 0.12 (1.95–2.35)	2.48 $\pm$ 0.15 (2.20–2.75)	2.32 $\pm$ 0.14 (2.05–2.55)
TD	1.08 $\pm$ 0.08 (0.95–1.22)	1.32 $\pm$ 0.10 (1.15–1.50)	1.20 $\pm$ 0.09 (1.05–1.38)
FDW	0.82 $\pm$ 0.06 (0.72–0.92)	0.98 $\pm$ 0.08 (0.84–1.12)	0.91 $\pm$ 0.07 (0.80–1.04)

4.4 Systematic Account: Formal Taxonomic Description

In accordance with the mandatory provisions of Articles 13 and 16 of the International Code of Zoological Nomenclature (International Commission on Zoological Nomenclature, 1999), we formally establish this distinct evolutionary lineage as a new nominal species.

Class: Amphibia Blainville, 1816

Order: Anura Fischer von Waldheim, 1813

Family: Dendrobatidae Cope, 1865

Genus: *Ranitomeya* Bauer, 1986

Ranitomeya aurata **sp. nov.** Langoo, 2026

Golden-Crowned Poison Frog / Ranita Venenosa Dorada

Holotype: QCAZ 45210, an adult male (SVL 16.52 mm), collected on 12 March 2023 by Sahil Langoo in primary submontane pluvial cloud forest along the western slopes of Cordillera Azul, Region San Martín, Peru.

Paratype: AMNH 128940 (adult female) and **Paratype:** MUSM 39140 (adult male), collected syntopically with holotype under authorization permit AUT-IFS-2023-088.

Type Locality: Cordillera Azul, San Martín, Peru (6°55'12"S, 76°11'45"W, 840 m elevation above sea level).

Diagnosis

Ranitomeya aurata **sp. nov.** is diagnosed by the following combination of characters: (1) small adult size (mean SVL 16.42 mm in males, 16.85 mm in females); (2) dorsum jet black with a distinctive, brilliant metallic golden-yellow cephalic hood extending posteriorly as fine dorsolateral stripes; (3) flank and limb ground color deep cobalt blue overlaid by coarse black circular reticulations; (4) gular and venter bright canary yellow with bold black ocelli; (5) teeth absent from maxillae and premaxillae; (6) Finger I distinctly shorter than Finger II; (7) digital discs moderately expanded; (8) distinct pairwise *COI* sequence divergence ($> 6.4\%$) and diagnostic substitutions at positions 112 (C→T) and 345 (A→G) of the Folmer alignment.

Differential Diagnosis

Ranitomeya aurata **sp. nov.** differs readily from its sister species *Ranitomeya fantastica* by its significantly smaller adult body size (SVL 15.60 —17.20 mm vs. 17.80 —20.40 mm in *R. fantastica*; $P < 0.001$, ANOVA, [Appendix C](#)), its gleaming metallic golden cephalic hood (versus red, orange, or uniform yellow in *R. fantastica*), and deep cobalt limb reticulations (versus charcoal or light turquoise). It is distinguished from *Ranitomeya summersi* by possessing dorsal striping and vibrant blue ventral patterning (lacking in

R. summersi).

Description of Holotype

Adult male in excellent preservation state (SVL 16.52 mm). Head narrower than body; head width 5.18 mm; head length 4.90 mm. Snout rounded in dorsal view, slightly protruding in lateral profile. Canthus rostralis rounded; loreal region concave. Nostrils directed posterolaterally. Eyes large and prominent, diameter 2.20 mm. Tympanum distinct, circular, diameter 1.10 mm, approximately 50% of eye diameter. Forelimbs slender; relative lengths of adpressed fingers: I < II < IV < III. Hindlimbs moderately robust; tibia length 7.45 mm; foot length 6.90 mm. Skin of dorsum, flanks, and venter smooth.

Etymology

The specific epithet *aurata* is a Latin adjective meaning “golden” or “gilded,” directly referencing the radiant, metallic golden cephalic hood and dorsolateral stripes that distinguish this species in life.

Chapter 5

Discussion and Evolutionary Synthesis

5.1 Integrative Taxonomy and Cryptic Speciation

The discovery and formal description of *Ranitomeya aurata* **sp. nov.** underscores the critical necessity of integrative taxonomy in hyperdiverse Neotropical ecosystems. While traditional taxonomy relied almost exclusively on external chromatic characters or gross skeletal morphology, our findings demonstrate that striking molecular divergence can remain obscured beneath superficial morphological convergence. As emphasized by Brown et al. (2011), dendrobatid frogs exhibit extreme color polytypy across continuous geographic clines, making it perilous to delimit taxa on coloration alone without genealogical corroboration.

By synthesizing multi-locus gene genealogies (*COI*, *16S*, and *RAG1*) with objective coalescent modeling (MPTP and BPP) and multivariate morphometrics, this study provides unequivocal evidence that *Ranitomeya aurata* constitutes an evolutionary lineage evolving independently from *Ranitomeya fantastica* sensu stricto. The concordant divergence observed across both mitochondrial (> 6.4%) and nuclear partitions demonstrates that reproductive isolation has persisted across substantial evolutionary time.

5.2 Historical Biogeography and Mechanisms of Diversification

The diversification of *Ranitomeya* within the sub-Andean Cordillera Oriental provides compelling empirical support for the combined influence of tectonic orogenesis and river-

ine vicariance (Santos et al., 2009; Avise, 2000). The Cordillera Azul forms an isolated, elevated anticlinal ridge situated east of the main Andean chain, bounded by the Huallaga and Ucayali river drainages.

$$T_{\text{div}} = \frac{d_{xy} - \frac{1}{2}(\pi_A + \pi_B)}{2\mu} \quad (5.1)$$

Applying molecular clock calibrations under the multispecies coalescent (Equation (5.1)), the divergence between *Ranitomeya aurata* and *Ranitomeya fantastica* is estimated at approximately 3.8 ± 0.6 Ma before present (late Pliocene). This temporal window aligns precisely with the major late Pliocene uplift pulse of the sub-Andean front-ranges, which redirected ancestral river systems and generated steep elevational and rainfall gradients. Subsequent Pleistocene climatic fluctuations likely reinforced lineage isolation by fragmenting continuous pluvial cloud forests into discrete montane refugia.

5.3 Reproductive Biology, Microhabitat, and Aposematic Ecology

Like other members of the *Ranitomeya fantastica* species group, *Ranitomeya aurata* exhibits obligate phytotelm breeding behavior. Field observations documented adult males transporting individual newly hatched tadpoles to rain-filled axils of epiphytic bromeliads (*Guzmania* and *Vriesea* spp.) positioned 2.5 m to 8.0 m above the forest floor.

The striking metallic golden cephalic coloration documented in *Ranitomeya aurata* represents an effective aposematic signal against avian and mammalian predators. Preliminary skin secretion screens indicate the presence of histrionicotoxin and pumiliotoxin alkaloid classes, which are acquired via specialized predation on formicine ants and oribatid soil mites. The retention of vivid cobalt limb reticulations may serve as disruptive camouflage against dappled rainforest light when stationary, while flashing warning coloration during active diurnal foraging.

5.4 Conservation Implications and Red List Assessment

Amphibians worldwide face unprecedented extinction risks driven by anthropogenic habitat degradation, chytridiomycosis outbreaks, and commercial exploitation. The known geographic distribution of *Ranitomeya aurata* is currently restricted to a narrow eleva-

tional belt (750 m–980 m) spanning less than 350 km² within the buffer zone of the Parque Nacional Cordillera Azul.

Satellite remote sensing data reveal accelerating rates of deforestation surrounding the type locality, driven by smallholder agricultural expansion, timber extraction, and infrastructure development. Furthermore, brightly colored dendrobatids are under severe collection pressure from the international illegal pet trade. In accordance with the official IUCN Red List Categories and Criteria, we recommend that *Ranitomeya aurata* be formally classified as:

Recommended IUCN Status: Endangered (EN) under Criteria B1ab(iii) + B2ab(iii)

Immediate conservation interventions should prioritize:

1. Establishing participatory community reserves in the buffer zones of Cordillera Azul.
2. Including *Ranitomeya aurata* on Appendix II of the Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES).
3. Implementing community-based ranger patrols to deter illegal poaching.

5.5 Conclusions and Future Trajectories

This doctoral dissertation establishes a reproducible, integrative taxonomic framework for resolving cryptic diversity in tropical anurans. Future investigations should incorporate ultraconserved elements (UCEs) or whole-genome resequencing to model historical gene flow across contact zones, alongside high-resolution bioacoustic recordings to decipher mate-recognition signaling divergence. The continued preservation of museum vouchers in public institutional repositories remains an irreplaceable foundation for documenting the biosphere during an era of rapid environmental change.

References

- Avice, J. C. (2000). *Phylogeography: The History and Formation of Species*. Cambridge, Massachusetts, USA: Harvard University Press.
- Brown, J. L., E. Twomey, A. Amézquita, M. B. De Souza, J. P. Caldwell, S. Lötters, R. von May, P. R. Melo-Sampaio, J. Janovec, P. Pérez-Peña, M. Pepper, and K. Summers (2011). “A taxonomic revision of the Neotropical poison frog genus *Ranitomeya* (Amphibia: Dendrobatidae)”. In: *Zootaxa* 3083.1, pp. 1–120. DOI: [10.11646/zootaxa.3083.1.1](https://doi.org/10.11646/zootaxa.3083.1.1).
- Darwin, C. (1859). *On the Origin of Species by Means of Natural Selection, or the Preservation of Favoured Races in the Struggle for Life*. London, United Kingdom: John Murray.
- Edgar, R. C. (2004). “MUSCLE: multiple sequence alignment with high accuracy and high throughput”. In: *Nucleic Acids Research* 32.5, pp. 1792–1797. DOI: [10.1093/nar/gkh340](https://doi.org/10.1093/nar/gkh340).
- Felsenstein, J. (1985). “Confidence Limits on Phylogenies: An Approach Using the Bootstrap”. In: *Evolution* 39.4, pp. 783–791. DOI: [10.1111/j.1558-5646.1985.tb00420.x](https://doi.org/10.1111/j.1558-5646.1985.tb00420.x).
- Folmer, O., M. Black, W. Hoeh, R. Lutz, and R. Vrijenhoek (1994). “DNA primers for amplification of mitochondrial cytochrome c oxidase subunit I from diverse metazoan invertebrates”. In: *Molecular Marine Biology and Biotechnology* 3.5, pp. 294–299.
- Hebert, P. D. N., A. Cywinska, S. L. Ball, and J. R. deWaard (2003). “Biological identifications through DNA barcodes”. In: *Proceedings of the Royal Society of London. Series B: Biological Sciences* 270.1512, pp. 313–321. DOI: [10.1098/rspb.2002.2218](https://doi.org/10.1098/rspb.2002.2218).
- Hennig, W. (1966). *Phylogenetic Systematics*. Urbana, Illinois, USA: University of Illinois Press.
- International Commission on Zoological Nomenclature (1999). *International Code of Zoological Nomenclature*. 4th. London, United Kingdom: The International Trust for Zoological Nomenclature.

- Linnaeus, C. (1758). *Systema Naturae per Regna Tria Naturae, Secundum Classes, Ordines, Genera, Species, cum Characteribus, Differentiis, Synonymis, Locis*. 10th. Holmiae (Stockholm), Sweden: Laurentii Salvii.
- Mayr, E. (1942). *Systematics and the Origin of Species from the Viewpoint of a Zoologist*. New York, USA: Columbia University Press.
- National Research Council (2011). *Guide for the Care and Use of Laboratory Animals*. 8th. Washington, DC, USA: The National Academies Press. DOI: [10.17226/12910](https://doi.org/10.17226/12910).
- Nguyen, L.-T., H. A. Schmidt, A. von Haeseler, and B. Q. Minh (2015). “IQ-TREE: a fast and effective stochastic algorithm for estimating maximum-likelihood phylogenies”. In: *Molecular Biology and Evolution* 32.1, pp. 268–274. DOI: [10.1093/molbev/msu300](https://doi.org/10.1093/molbev/msu300).
- Rannala, B. and Z. Yang (2003). “Bayes estimation of species divergence times and ancestral population sizes using DNA sequences from multiple loci”. In: *Genetics* 164.4, pp. 1645–1656. DOI: [10.1093/genetics/164.4.1645](https://doi.org/10.1093/genetics/164.4.1645).
- Ronquist, F. and J. P. Huelsenbeck (2003). “MrBayes 3: Bayesian phylogenetic inference under mixed models”. In: *Bioinformatics* 19.12, pp. 1572–1574. DOI: [10.1093/bioinformatics/btg180](https://doi.org/10.1093/bioinformatics/btg180).
- Sambrook, J. and D. W. Russell (2001). *Molecular Cloning: A Laboratory Manual*. 3rd. Cold Spring Harbor, New York, USA: Cold Spring Harbor Laboratory Press.
- Santos, J. C., L. A. Coloma, K. Summers, J. P. Caldwell, R. Ree, and D. C. Cannatella (2009). “Amazonian amphibian diversity is primarily derived from late Miocene Andean orogeny”. In: *PLoS Biology* 7.3, e1000056. DOI: [10.1371/journal.pbio.1000056](https://doi.org/10.1371/journal.pbio.1000056).
- Zhang, J., P. Kapli, P. Pavlidis, and A. Stamatakis (2013). “A general species delimitation method with applications to temperate and tropical biotas”. In: *Bioinformatics* 29.22, pp. 2869–2876. DOI: [10.1093/bioinformatics/btt499](https://doi.org/10.1093/bioinformatics/btt499).

Appendix A

Examined Specimen Catalog and Georeferenced Localities

This appendix provides catalog metadata for representative voucher specimens examined in this study. Institutional acronyms follow standard museum codens: AMNH = American Museum of Natural History, New York; MUSM = Museo de Historia Natural de la Universidad Nacional Mayor de San Marcos, Lima; QCAZ = Museo de Zoología, Pontificia Universidad Católica del Ecuador, Quito; USNM = National Museum of Natural History, Smithsonian Institution, Washington, DC.

Catalog No.	Taxon		Locality & Country	Coordinates (WGS84)	GenBank <i>COI</i>
QCAZ 45210*	<i>Ranitomeya</i> <i>rata</i> sp. nov.	<i>au-</i>	Cordillera Azul, San Martín, Peru	6°55'12"S, 76°11'45"W	OQ849101
AMNH 128940**	<i>Ranitomeya</i> <i>rata</i> sp. nov.	<i>au-</i>	Cordillera Azul, San Martín, Peru	6°55'10"S, 76°11'50"W	OQ849102
MUSM 39140**	<i>Ranitomeya</i> <i>rata</i> sp. nov.	<i>au-</i>	Cordillera Azul, San Martín, Peru	6°55'18"S, 76°11'40"W	OQ849103
MUSM 39141	<i>Ranitomeya</i> <i>rata</i> sp. nov.	<i>au-</i>	Pampa Hermosa, San Martín, Peru	6°58'2"S, 76°14'12"W	OQ849104
MUSM 39142	<i>Ranitomeya</i> <i>rata</i> sp. nov.	<i>au-</i>	Pampa Hermosa, San Martín, Peru	6°58'15"S, 76°14'20"W	OQ849105
QCAZ 41800	<i>Ranitomeya</i> <i>fantastica</i>	<i>fantas-</i>	Tarapoto, San Martín, Peru	6°29'30"S, 76°22'15"W	HQ840210

Catalog No.	Taxon	Locality & Country	Coordinates (WGS84)	GenBank <i>COI</i>
QCAZ 41801	<i>Ranitomeya fantas- tica</i>	Tarapoto, San Martín, Peru	6°29'35"S, 76°22'18"W	HQ840211
AMNH 115620	<i>Ranitomeya fantas- tica</i>	Cainarachi, San Martín, Peru	6°21'45"S, 76°17'30"W	HQ840212
AMNH 115621	<i>Ranitomeya fantas- tica</i>	Cainarachi, San Martín, Peru	6°21'50"S, 76°17'25"W	HQ840213
USNM 538200	<i>Ranitomeya sum- mersi</i>	Sauce, San Martín, Peru	6°41'12"S, 76°13'45"W	HQ840245
USNM 538201	<i>Ranitomeya sum- mersi</i>	Sauce, San Martín, Peru	6°41'15"S, 76°13'50"W	HQ840246
MUSM 27150	<i>Ranitomeya sum- mersi</i>	Chazuta, San Martín, Peru	6°34'22"S, 76°8'10"W	HQ840247
QCAZ 32110	<i>Ranitomeya vari- abilis</i>	Yasuní National Park, Orellana, Ecuador	0°40'25"S, 76°23'50"W	DQ283100
QCAZ 32111	<i>Ranitomeya vari- abilis</i>	Yasuní National Park, Orellana, Ecuador	0°40'30"S, 76°23'45"W	DQ283101
AMNH 98720	<i>Ranitomeya amazon- ica</i>	Iquitos, Loreto, Peru	3°45'0"S, 73°15'0"W	EU342115
AMNH 98721	<i>Ranitomeya amazon- ica</i>	Iquitos, Loreto, Peru	3°45'10"S, 73°15'12"W	EU342116
USNM 342100	<i>Dendrobates tincto- rius</i>	Camp Patawa, French Guiana	4°32'40"N, 52°19'15"W	AY843201

* Denotes Holotype voucher; ** Denotes Paratype voucher.

Appendix B

Oligonucleotide Primers and Amplification Conditions

This appendix documents all polymerase chain reaction (PCR) oligonucleotide primers used for targeted Sanger sequencing in this dissertation. All oligonucleotides were synthesized via standard solid-phase phosphoramidite chemistry and purified via reverse-phase cartridge desalting.

Table B.1: Oligonucleotide primers utilized for PCR amplification and bidirectional Sanger sequencing.

Locus	Primer	Sequence (5' → 3')	T_a (°C)	Size (bp)	Source
<i>COI</i>	LCO1490	GGTCAACAAATCATAAAGATATTGG	50	658	Folmer et al. (1994)
<i>COI</i>	HCO2198	TAACTTCAGGGTGACCAAAAAATCA	50	658	Folmer et al. (1994)
<i>16S rRNA</i>	16Sar-L	CGCCTGTTTATCAAAAACAT	52	550	Palumbi (1996)
<i>16S rRNA</i>	16Sbr-H	CCGGTCTGAACTCAGATCACGT	52	550	Palumbi (1996)
<i>RAG1</i>	RAG1-F1	AGCTGCAGYCARTAYCAYAARATGT	56	820	Santos et al. (2009)
<i>RAG1</i>	RAG1-R1	AACTCAGCTGCYTCRTTRAAYTC	56	820	Santos et al. (2009)

Cycling Profile Schematics

- Mitochondrial *COI*:

1. Initial denaturation: 94 °C for 180 s.

2. 35 cycles: Denaturation at 94 °C for 30 s; Annealing at 50 °C for 45 s; Elongation at 72 °C for 60 s.
3. Final extension: 72 °C for 420 s.
4. Soak: 4 °C hold.

- **Mitochondrial *16S rRNA*:**

1. Initial denaturation: 94 °C for 120 s.
2. 32 cycles: Denaturation at 94 °C for 30 s; Annealing at 52 °C for 40 s; Elongation at 72 °C for 50 s.
3. Final extension: 72 °C for 300 s.

- **Nuclear Exon *RAG1* (Touchdown PCR):**

1. Initial denaturation: 95 °C for 300 s.
2. 10 touchdown cycles: 94 °C for 45 s, 60 °C to 51 °C (−1 °C per cycle) for 45 s, 72 °C for 90 s.
3. 28 standard cycles: 94 °C for 45 s, 55 °C for 45 s, 72 °C for 90 s.
4. Final extension: 72 °C for 600 s.

Appendix C

Supplementary Statistical Models and Morphometric Analyses

This appendix presents detailed statistical testing outputs, including Analysis of Variance (ANOVA) tables, General Linear Model (GLM) parameters, and Akaike Information Criterion corrected for small sample sizes (AICc) for morphological traits across the *Ranitomeya fantastica* species group.

C.1 Factorial ANOVA on Morphometric Traits

A two-way factorial ANOVA was executed to assess the fixed effects of Taxon (*Ranitomeya aurata* **sp. nov.**, *Ranitomeya fantastica*, and *Ranitomeya summersi*), Sex (Male vs. Female), and their interaction on adult Snout-Vent Length (SVL) and Head Width (HW).

Table C.1: Two-way factorial ANOVA results evaluating variation in adult Snout-Vent Length (SVL) across taxa and sexes.

Source of Variation	df	Sum of Sq	Mean Sq	<i>F</i> -statistic	<i>P</i> -value
Taxon	2	78.412	39.206	134.72	< 0.0001
Sex	1	4.285	4.285	14.72	0.0002
Taxon × Sex	2	0.418	0.209	0.72	0.4901
Residuals (Error)	68	19.791	0.291	–	–
Total	73	102.906	–	–	–

C.2 Tukey’s Honestly Significant Difference (HSD) Post-Hoc Tests

Pairwise post-hoc comparisons with Tukey’s adjustment demonstrated that adult *Ranitomeya aurata* **sp. nov.** is significantly smaller in overall SVL than both *Ranitomeya fantastica* (difference: -2.52 mm; 95% CI:

Table C.2: Two-way factorial ANOVA results evaluating variation in adult Head Width (HW) across taxa and sexes.

Source of Variation	df	Sum of Sq	Mean Sq	<i>F</i> -statistic	<i>P</i> -value
Taxon	2	8.924	4.462	71.97	< 0.0001
Sex	1	0.312	0.312	5.03	0.0282
Taxon $\times$ Sex	2	0.084	0.042	0.68	0.5102
Residuals (Error)	68	4.216	0.062	–	–
Total	73	13.536	–	–	–

–2.86 to –2.18; $P < 0.0001$) and *Ranitomeya summersi* (difference: –1.40 mm; 95% CI: –1.76 to –1.04; $P < 0.0001$).

C.3 AICc Model Selection: Biogeographical Drivers of Body Size

Candidate linear models evaluating the predictive power of Elevation (m a.s.l.), River Drainage Basin, and Mean Annual Precipitation (Bio12 from WorldClim v2.1) in explaining adult body size (SVL) were evaluated using Akaike Information Criterion corrected for small sample size (AICc).

Table C.3: Model selection statistics for candidate regression models explaining body size (SVL) variation across sub-Andean dendrobatids.

Model Formulation	<i>K</i>	$\log(L)$	AICc	ΔAICc	w_i
SVL $\sim$ Taxon + Elevation	4	-61.24	131.06	0.00	0.682
SVL $\sim$ Taxon + Elevation + Basin	6	-59.88	133.02	1.96	0.256
SVL $\sim$ Taxon	3	-65.10	136.54	5.48	0.044
SVL $\sim$ Elevation + Precipitation	4	-78.40	165.38	34.32	< 0.001
SVL $\sim$ Null (Intercept only)	2	-112.50	229.17	98.11	< 0.001

Note: K is the number of estimated model parameters; $\log(L)$ is the log-likelihood; $\Delta\text{AICc} = \text{AICc}_i - \min(\text{AICc})$; w_i is the Akaike model weight.